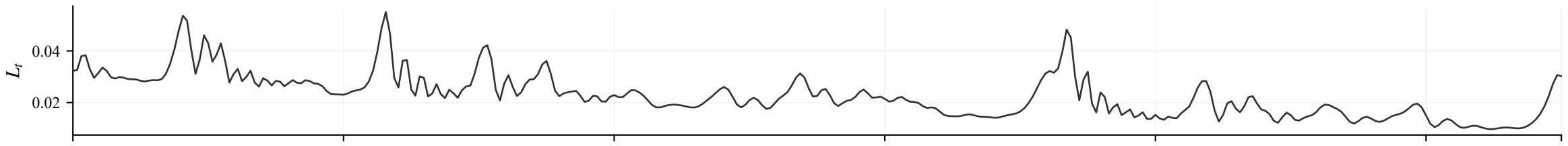
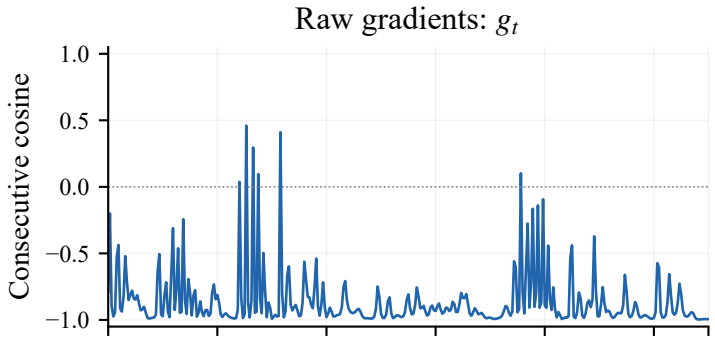
CIFAR-10 MLP: raw-gradient, output-change and hidden-change SVDs

Same 353 states | All 5,000 examples | Fixed uncentered basis fitted independently in each space

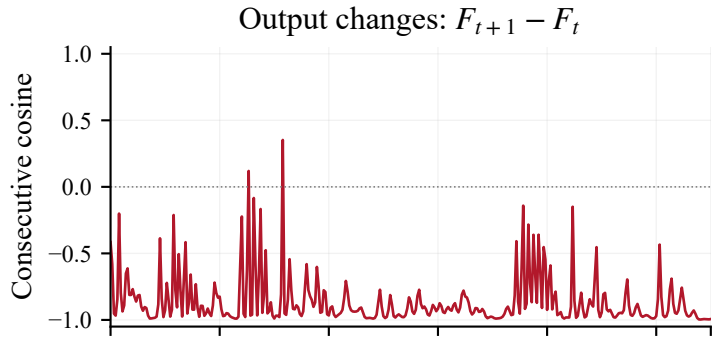
(a) Full training loss: the same trajectory for all three spaces



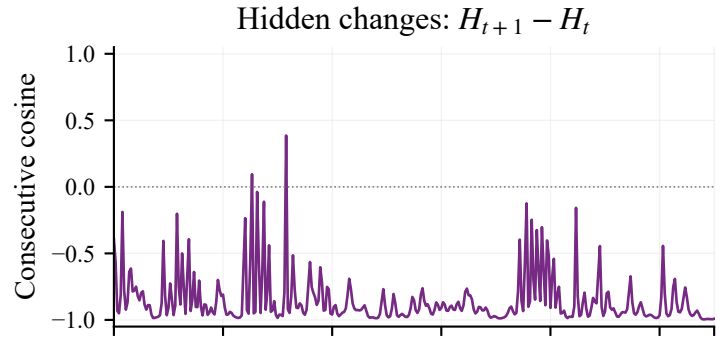
Raw gradients: g_t



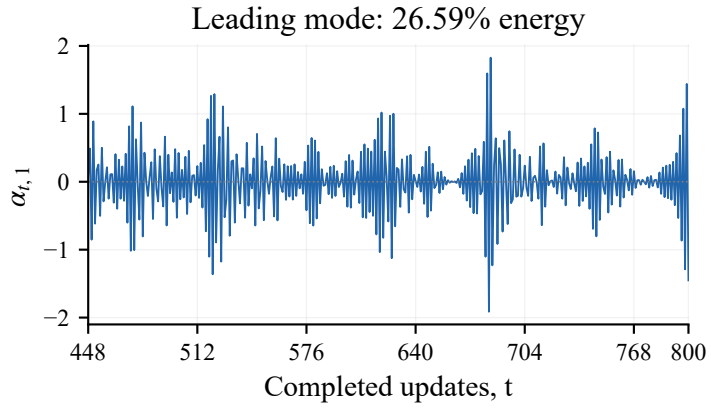
Output changes: $F_{t+1} - F_t$



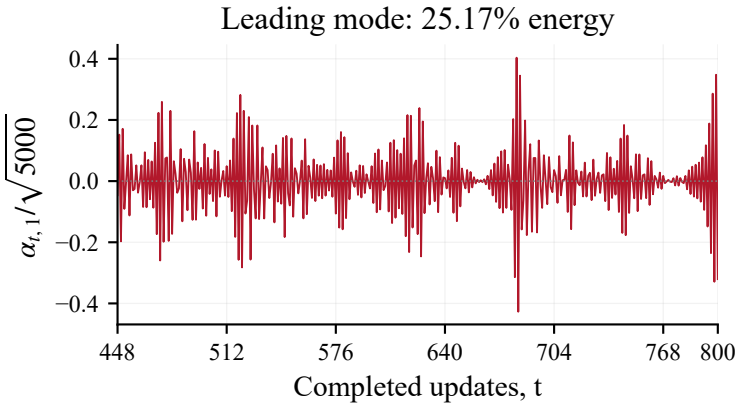
Hidden changes: $H_{t+1} - H_t$



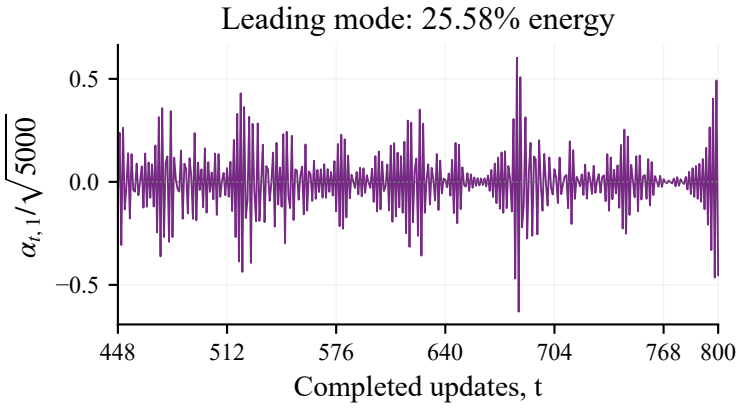
Leading mode: 26.59% energy



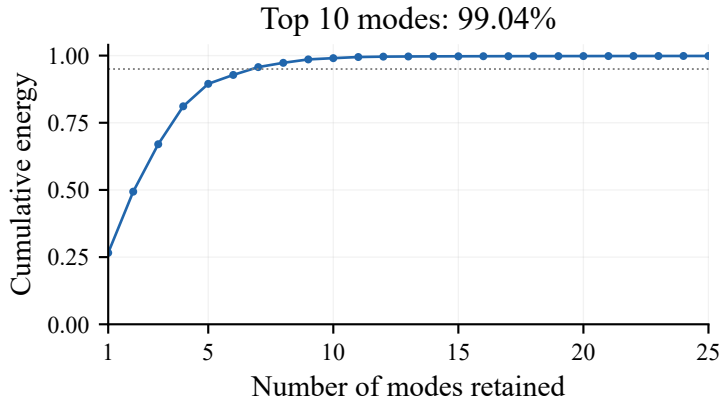
Leading mode: 25.17% energy



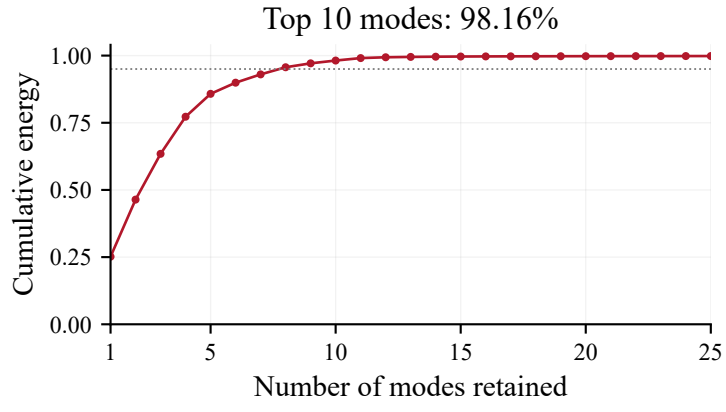
Leading mode: 25.58% energy



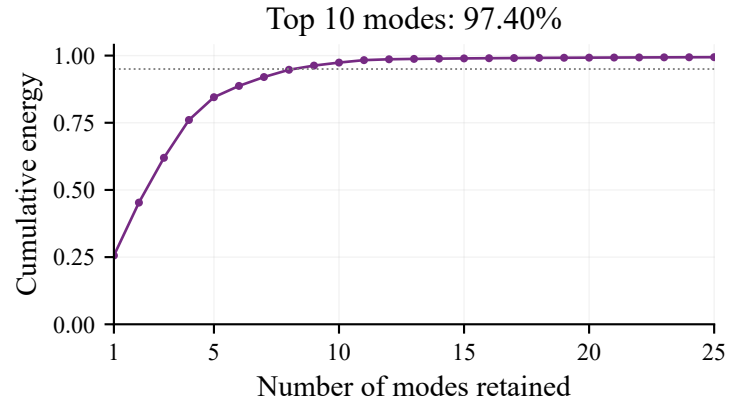
Top 10 modes: 99.04%



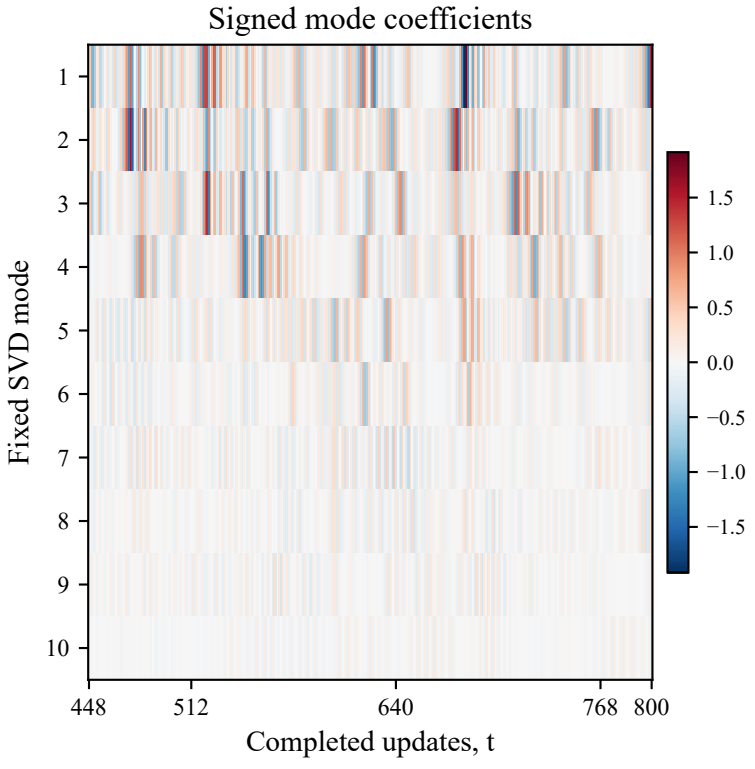
Top 10 modes: 98.16%



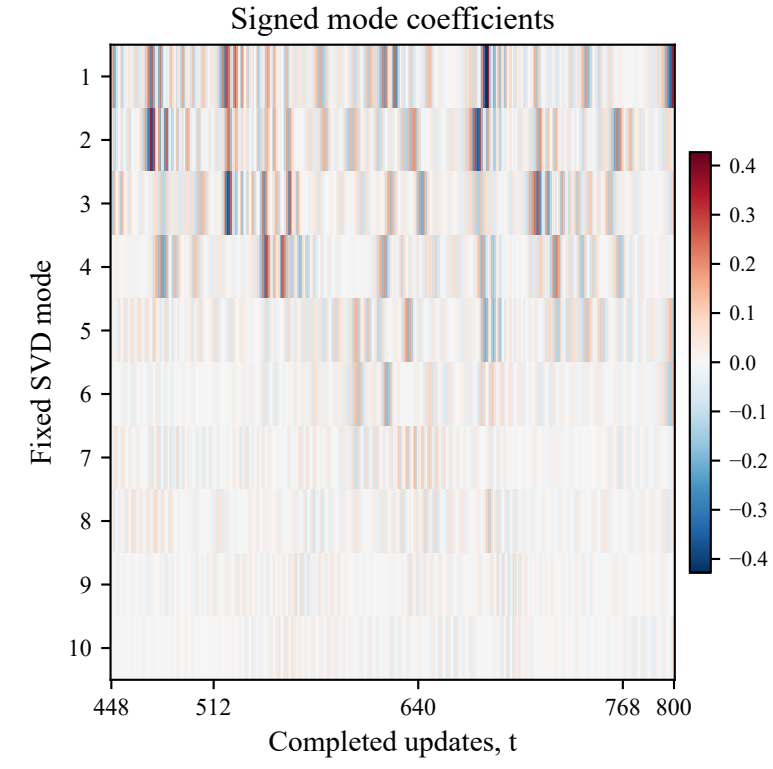
Top 10 modes: 97.40%



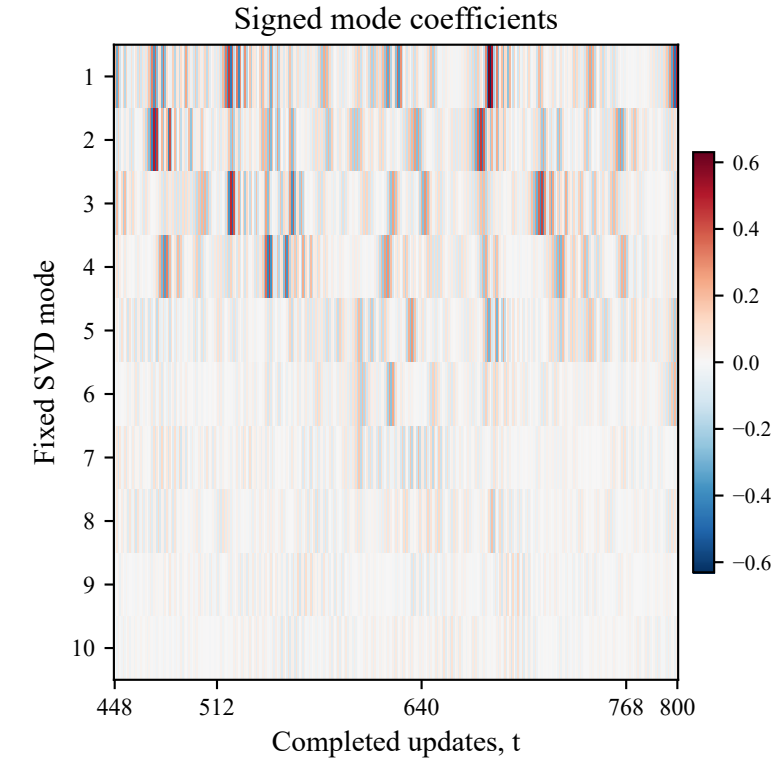
Signed mode coefficients



Signed mode coefficients



Signed mode coefficients



Gradient alignment: $\cos(g_t, g_{t-1})$. Output/hidden alignment: incoming versus outgoing displacement around t.

Gradient rows are raw gradients, NOT differences. Hidden rows use the second tanh, immediately before the linear head.

Mode signs are fixed within each space. Mode numbers and colors do not imply corresponding axes across spaces.